Projet Graph Analysis

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```r
library(igraph)
library(sna)
library(readr)
library(dplyr)
library(ggplot2)
library(tidygraph)
library(ggraph)
library(grid)
library(knitr)

# Data
edges <- read_csv("spurs_assist_network_edges_2024_25_regular_season.csv", show_col_types = FALSE)
followers <- read_csv("spurs_instagram_followers_simple.csv", show_col_types = FALSE)
```

Question 3

```r
# Keep only real ties
edges <- edges %>%
  select(from_player, to_player, assists) %>%
  filter(assists > 0)

spurs_net <- graph_from_data_frame(
  d = edges,
  directed = TRUE,
  vertices = followers %>% rename(name = player)
)

E(spurs_net)$weight <- E(spurs_net)$assists
V(spurs_net)$instagram_followers <- followers$instagram_followers[
  match(V(spurs_net)$name, followers$player)
]
```

Question 4

```r
cat("Number of nodes:", vcount(spurs_net), "\n")
```

```
## Number of nodes: 20
```

```r
cat("Number of edges:", ecount(spurs_net), "\n")
```

```
## Number of edges: 226
```

```r
cat("Directed graph:", is_directed(spurs_net), "\n")
```

```
## Directed graph: TRUE
```

```r
cat("Weighted graph:", is_weighted(spurs_net), "\n")
```

```
## Weighted graph: TRUE
```

```r
cat("\nVertex attributes:\n")
```

```
##
## Vertex attributes:
```

```r
print(vertex_attr_names(spurs_net))
```

```
## [1] "name"                "instagram_followers"
```

```r
cat("\nEdge attributes:\n")
```

```
##
## Edge attributes:
```

```r
print(edge_attr_names(spurs_net))
```

```
## [1] "assists" "weight"
```

```r
density_spurs <- edge_density(spurs_net, loops = FALSE)
reciprocity_spurs <- reciprocity(spurs_net)

spurs_net_und <- as_undirected(
  spurs_net,
  mode = "collapse",
  edge.attr.comb = list(weight = "sum")
)

transitivity_spurs <- transitivity(spurs_net_und, type = "global")

network_stats <- data.frame(
  Statistic = c("Density", "Reciprocity", "Transitivity"),
  Value = c(density_spurs, reciprocity_spurs, transitivity_spurs)
)

network_stats
```

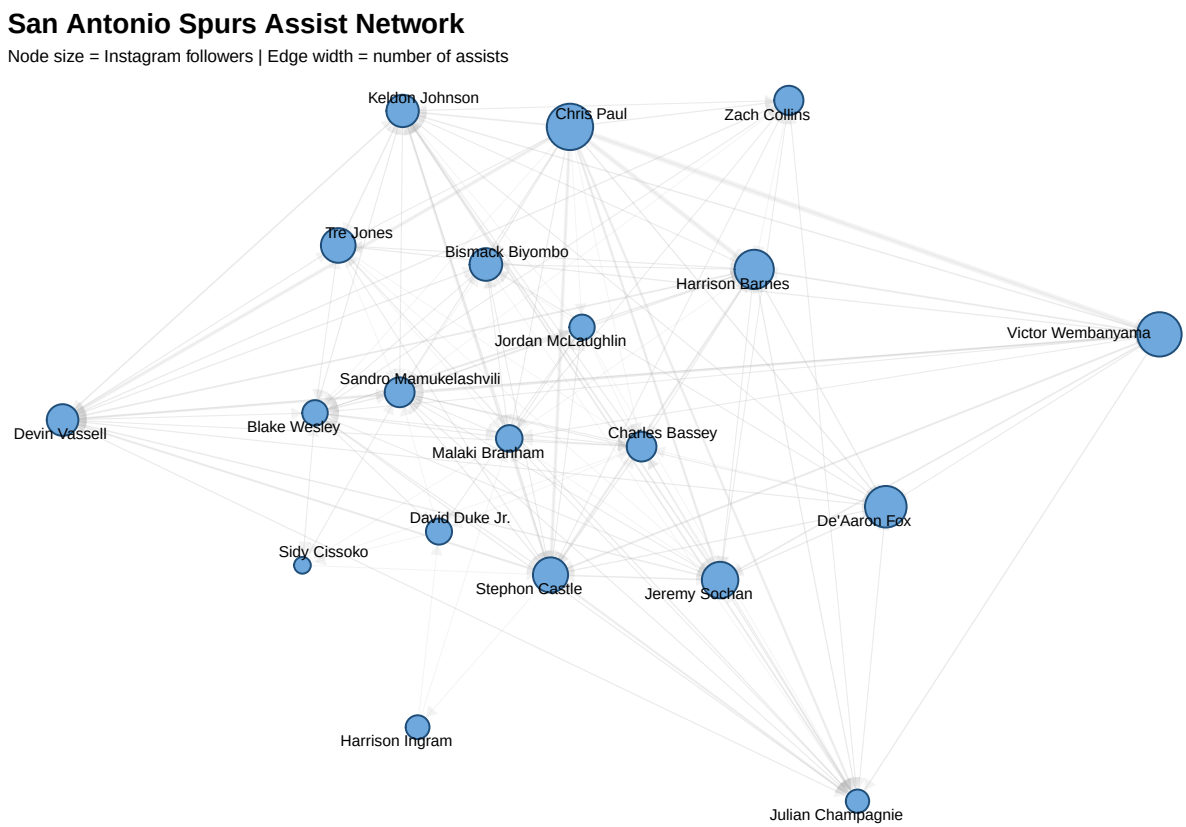
```
##      Statistic     Value
## 1      Density 0.5947368
## 2  Reciprocity 0.8672566
## 3 Transitivity 0.8004765
```

```r
attribute_summary <- data.frame(
  Min_Followers = min(V(spurs_net)$instagram_followers, na.rm = TRUE),
  Median_Followers = median(V(spurs_net)$instagram_followers, na.rm = TRUE),
  Mean_Followers = mean(V(spurs_net)$instagram_followers, na.rm = TRUE),
  Max_Followers = max(V(spurs_net)$instagram_followers, na.rm = TRUE)
)

attribute_summary
```

```
##   Min_Followers Median_Followers Mean_Followers Max_Followers
## 1         33000           124500        1092700      11400000
```

```r
set.seed(123)

graph_tbl <- as_tbl_graph(spurs_net) %>%
  activate(nodes) %>%
  mutate(followers_log = log1p(instagram_followers))

ggraph(graph_tbl, layout = "kk") +
  geom_edge_link(
    aes(width = weight),
    colour = "grey70",
    alpha = 0.15,
    arrow = arrow(length = unit(2.2, "mm"), type = "closed"),
    end_cap = circle(2.8, "mm"),
    start_cap = circle(1.2, "mm")
  ) +
  scale_edge_width(range = c(0.2, 1.5), guide = "none") +
  geom_node_point(
    aes(size = followers_log),
    shape = 21,
    fill = "#6fa8dc",
    colour = "#1f4e79",
    stroke = 0.8
  ) +
  scale_size_continuous(range = c(4, 12), guide = "none") +
  geom_node_text(
    aes(label = name),
    repel = TRUE,
    size = 3.0,
    colour = "black",
    point.padding = unit(0.15, "lines"),
    box.padding = unit(0.25, "lines")
  ) +
  labs(
    title = "San Antonio Spurs Assist Network",
    subtitle = "Node size = Instagram followers | Edge width = number of assists"
  ) +
```

```r
  theme_void() +
  theme(
    plot.title = element_text(size = 16, face = "bold", hjust = 0),
    plot.subtitle = element_text(size = 10, hjust = 0),
    plot.margin = margin(12, 12, 12, 12),
    plot.background = element_rect(fill = "white", colour = NA)
  )
```



Question 5

QAP

```r
# Binary adjacency matrix
spurs_adj <- as.matrix(as_adjacency_matrix(spurs_net, attr = NULL))
spurs_adj[spurs_adj > 0] <- 1

# Order followers like the adjacency matrix
followers_ordered <- followers %>%
  slice(match(rownames(spurs_adj), player))

followers_vec <- followers_ordered$instagram_followers
```

```r
# Receiver popularity matrix: entry (i,j) = followers of player j
pop_receiver <- matrix(
  rep(followers_vec, each = length(followers_vec)),
  nrow = length(followers_vec),
  byrow = FALSE
)

rownames(pop_receiver) <- rownames(spurs_adj)
colnames(pop_receiver) <- colnames(spurs_adj)

# Logistic QAP regression on directed graph
set.seed(123)
qap_model <- netlogit(
  spurs_adj,
  list(pop_receiver),
  mode = "digraph",
  nullhyp = "qapy",
  reps = 1000
)

summary(qap_model)
```

```
##
## Network Logit Model
##
## Coefficients:
##             Estimate     Exp(b)  Pr(<=b) Pr(>=b) Pr(>=|b|)
## (intercept) 3.368008e-01 1.40046 0.374   0.626   0.626
## x1          4.431238e-08 1.00000 0.626   0.374   0.677
##
## Goodness of Fit Statistics:
##
## Null deviance: 526.7919 on 380 degrees of freedom
## Residual deviance: 511.8908 on 378 degrees of freedom
## Chi-Squared test of fit improvement:
##   14.9011 on 2 degrees of freedom, p-value 0.000581121
## AIC: 515.8908    BIC: 523.7711
## Pseudo-R^2 Measures:
##  (Dn-Dr)/(Dn-Dr+dfn): 0.03773376
##  (Dn-Dr)/Dn: 0.02828651
## Contingency Table (predicted (rows) x actual (cols)):
##
##       0     1
## 0     0     0
## 1   154   226
##
##  Total Fraction Correct: 0.5947368
##  Fraction Predicted 1s Correct: 0.5947368
##  Fraction Predicted 0s Correct: NaN
##  False Negative Rate: 0
##  False Positive Rate: 1
##
## Test Diagnostics:
```

```
##
##  Null Hypothesis: qapy
##  Replications: 1000
##  Distribution Summary:
##
##        (intercept)       x1
## Min        1.42346 -4.68996
## 1stQ       2.74181 -1.59314
## Median     3.27165  0.32097
## Mean       3.39289 -0.04239
## 3rdQ       3.98737  1.64667
## Max        5.92947  3.48058
```

Communities

```r
spurs_net_und <- as_undirected(
  spurs_net,
  mode = "collapse",
  edge.attr.comb = list(weight = "sum")
)

# Louvain
set.seed(123)
spurs_louvain <- cluster_louvain(
  spurs_net_und,
  weights = E(spurs_net_und)$weight
)

membership(spurs_louvain)
```

```
##            Chris Paul       Bismack Biyombo       Harrison Barnes
##                     1                     1                     1
##          De'Aaron Fox          Zach Collins     Jordan McLaughlin
##                     1                     2                     2
##        Keldon Johnson        Charles Bassey         Devin Vassell
##                     2                     2                     1
##             Tre Jones        David Duke Jr. Sandro Mamukelashvili
##                     2                     2                     2
##     Julian Champagnie        Malaki Branham          Blake Wesley
##                     2                     2                     2
##         Jeremy Sochan       Harrison Ingram          Sidy Cissoko
##                     1                     2                     2
##     Victor Wembanyama        Stephon Castle
##                     1                     2
```

```r
length(unique(membership(spurs_louvain)))
```

```
## [1] 2
```

```r
sizes(spurs_louvain)
```

```
## Community sizes
##  1  2
##  7 13
```

```r
mod_score <- modularity(spurs_louvain)
mod_score
```

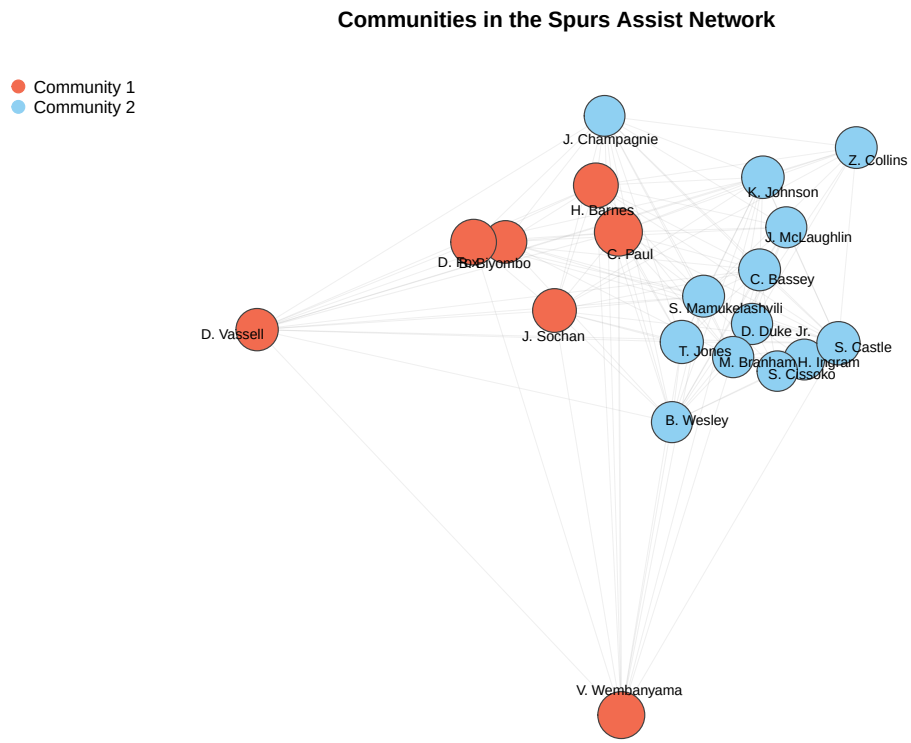
```
## [1] 0.0963635
```

```r
community_members <- data.frame(
  player = V(spurs_net_und)$name,
  community = membership(spurs_louvain),
  followers = V(spurs_net_und)$instagram_followers
)

community_members[order(community_members$community), ]
```

```
##                                     player community followers
## Chris Paul                       Chris Paul         1  11400000
## Bismack Biyombo             Bismack Biyombo         1    179000
## Harrison Barnes             Harrison Barnes         1    973000
## De'Aaron Fox                   De'Aaron Fox         1   2000000
## Devin Vassell                 Devin Vassell         1    145000
## Jeremy Sochan                 Jeremy Sochan         1    443000
## Victor Wembanyama         Victor Wembanyama         1   5300000
## Zach Collins                   Zach Collins         2     93000
## Jordan McLaughlin         Jordan McLaughlin         2     54000
## Keldon Johnson               Keldon Johnson         2    168000
## Charles Bassey               Charles Bassey         2    101000
## Tre Jones                         Tre Jones         2    280000
## David Duke Jr.               David Duke Jr.         2     59000
## Sandro Mamukelashvili Sandro Mamukelashvili         2    104000
## Julian Champagnie         Julian Champagnie         2     44000
## Malaki Branham               Malaki Branham         2     63000
## Blake Wesley                   Blake Wesley         2     55000
## Harrison Ingram             Harrison Ingram         2     46000
## Sidy Cissoko                   Sidy Cissoko         2     33000
## Stephon Castle               Stephon Castle         2    314000
```

```r
comm <- membership(spurs_louvain)

lay <- layout_with_kk(spurs_net_und)
lay <- norm_coords(lay, xmin = -1, xmax = 1, ymin = -1, ymax = 1)

# Separate communities a bit more
lay[comm == 1, 1] <- lay[comm == 1, 1] - 0.60
lay[comm == 2, 1] <- lay[comm == 2, 1] + 0.60

vertex_colors <- ifelse(comm == 1, "#F26B4F", "#8FD0F2")
```

```r
followers_log <- log1p(V(spurs_net_und)$instagram_followers)
vertex_sizes <- 9 + 7 * followers_log / max(followers_log)

edge_widths <- 0.2 + 1.2 * E(spurs_net_und)$weight / max(E(spurs_net_und)$weight)
edge_cols <- adjustcolor("grey50", alpha.f = 0.12)

# Shorter labels: "Victor Wembanyama" -> "V. Wembanyama"
short_labels <- sapply(strsplit(V(spurs_net_und)$name, " "), function(x) {
  if (length(x) >= 2) paste0(substr(x[1], 1, 1), ". ", paste(x[-1], collapse = " "))
  else x[1]
})

label_angle <- atan2(lay[, 2], lay[, 1])
label_dist <- 1.15

plot(
  spurs_net_und,
  layout = lay,
  vertex.color = vertex_colors,
  vertex.size = vertex_sizes,
  vertex.frame.color = "grey25",
  vertex.label = short_labels,
  vertex.label.cex = 0.72,
  vertex.label.color = "black",
  vertex.label.family = "sans",
  vertex.label.degree = label_angle,
  vertex.label.dist = label_dist,
  edge.width = edge_widths,
  edge.color = edge_cols,
  main = "Communities in the Spurs Assist Network"
)

legend(
  "topleft",
  legend = c("Community 1", "Community 2"),
  col = c("#F26B4F", "#8FD0F2"),
  pch = 19,
  pt.cex = 1.5,
  bty = "n",
  cex = 0.9
)
```



Statistics

```r
V(spurs_net_und)$community <- membership(spurs_louvain)

# Top 25%
top25_cutoff <- quantile(
  V(spurs_net_und)$instagram_followers,
  0.75,
  na.rm = TRUE
)

community_stats <- data.frame(
  player = V(spurs_net_und)$name,
  community = V(spurs_net_und)$community,
  followers = V(spurs_net_und)$instagram_followers
) %>%
  group_by(community) %>%
  summarise(
    n_players = n(),
    mean_followers = round(mean(followers, na.rm = TRUE), 0),
    median_followers = round(median(followers, na.rm = TRUE), 0),
    share_top25 = round(100 * mean(followers >= top25_cutoff, na.rm = TRUE), 1),
```

```r
    .groups = "drop"
  ) %>%
  mutate(
    community = paste("Community", community),
    mean_followers = format(mean_followers, big.mark = ",", scientific = FALSE),
    median_followers = format(median_followers, big.mark = ",", scientific = FALSE),
    share_top25 = paste0(share_top25, "%")
  )

kable(
  community_stats,
  col.names = c("Community", "Players", "Mean followers", "Median followers", "Top 25% share"),
  caption = "Instagram popularity by community",
  align = "c"
)
```

Table 1: Instagram popularity by community

Community	Players	Mean followers	Median followers	Top 25% share
Community 1	7	2,920,000	973,000	71.4%
Community 2	13	108,769	63,000	0%

Question 6

```r
additional_df <- data.frame(
  player = V(spurs_net)$name,
  followers = V(spurs_net)$instagram_followers,
  indegree_binary = igraph::degree(spurs_net, mode = "in"),
  assists_received = igraph::strength(spurs_net, mode = "in", weights = E(spurs_net)$weight)
)

additional_df <- additional_df[order(-additional_df$assists_received), ]
print(additional_df)
```

```
##                                     player followers indegree_binary
## Victor Wembanyama         Victor Wembanyama   5300000              12
## Harrison Barnes             Harrison Barnes    973000              16
## Devin Vassell                 Devin Vassell    145000              13
## Keldon Johnson               Keldon Johnson    168000              15
## Julian Champagnie         Julian Champagnie     44000              15
## Stephon Castle               Stephon Castle    314000              17
## Jeremy Sochan                 Jeremy Sochan    443000              13
## Sandro Mamukelashvili Sandro Mamukelashvili    104000              17
## Chris Paul                       Chris Paul  11400000              13
## Malaki Branham               Malaki Branham     63000              13
## Blake Wesley                   Blake Wesley     55000              17
## De'Aaron Fox                   De'Aaron Fox   2000000              10
## Bismack Biyombo             Bismack Biyombo    179000              10
## Zach Collins                   Zach Collins     93000               8
```

```
## Charles Bassey               Charles Bassey    101000              14
## Tre Jones                         Tre Jones    280000               8
## Jordan McLaughlin         Jordan McLaughlin     54000               6
## Sidy Cissoko                   Sidy Cissoko     33000               4
## David Duke Jr.               David Duke Jr.     59000               4
## Harrison Ingram             Harrison Ingram     46000               1
##                       assists_received
## Victor Wembanyama                  309
## Harrison Barnes                    265
## Devin Vassell                      265
## Keldon Johnson                     256
## Julian Champagnie                  245
## Stephon Castle                     226
## Jeremy Sochan                      193
## Sandro Mamukelashvili              114
## Chris Paul                          93
## Malaki Branham                      59
## Blake Wesley                        58
## De'Aaron Fox                        57
## Bismack Biyombo                     53
## Zach Collins                        53
## Charles Bassey                      53
## Tre Jones                           26
## Jordan McLaughlin                   10
## Sidy Cissoko                         8
## David Duke Jr.                       4
## Harrison Ingram                      1
```

```r
# Spearman correlations
cor_binary <- cor.test(
  additional_df$followers,
  additional_df$indegree_binary,
  method = "spearman",
  exact = FALSE
)

cor_weighted <- cor.test(
  additional_df$followers,
  additional_df$assists_received,
  method = "spearman",
  exact = FALSE
)

print(cor_binary)
```

```
##
##  Spearman's rank correlation rho
##
## data:  additional_df$followers and additional_df$indegree_binary
## S = 969.55, p-value = 0.2478
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##       rho
## 0.2710131
```

```r
print(cor_weighted)
```

```
##
##  Spearman's rank correlation rho
##
## data:  additional_df$followers and additional_df$assists_received
## S = 626.68, p-value = 0.01652
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##       rho
## 0.5288145
```

```r
set.seed(123)

obs_cor <- cor(
  additional_df$followers,
  additional_df$assists_received,
  method = "spearman"
)

perm_corr <- replicate(
  5000,
  cor(sample(additional_df$followers), additional_df$assists_received, method = "spearman")
)

p_one_sided <- mean(perm_corr >= obs_cor)
p_two_sided <- mean(abs(perm_corr) >= abs(obs_cor))

cat("Observed Spearman correlation:", round(obs_cor, 3), "\n")
```

```
## Observed Spearman correlation: 0.529
```

```r
cat("Empirical one-sided p-value:", round(p_one_sided, 3), "\n")
```

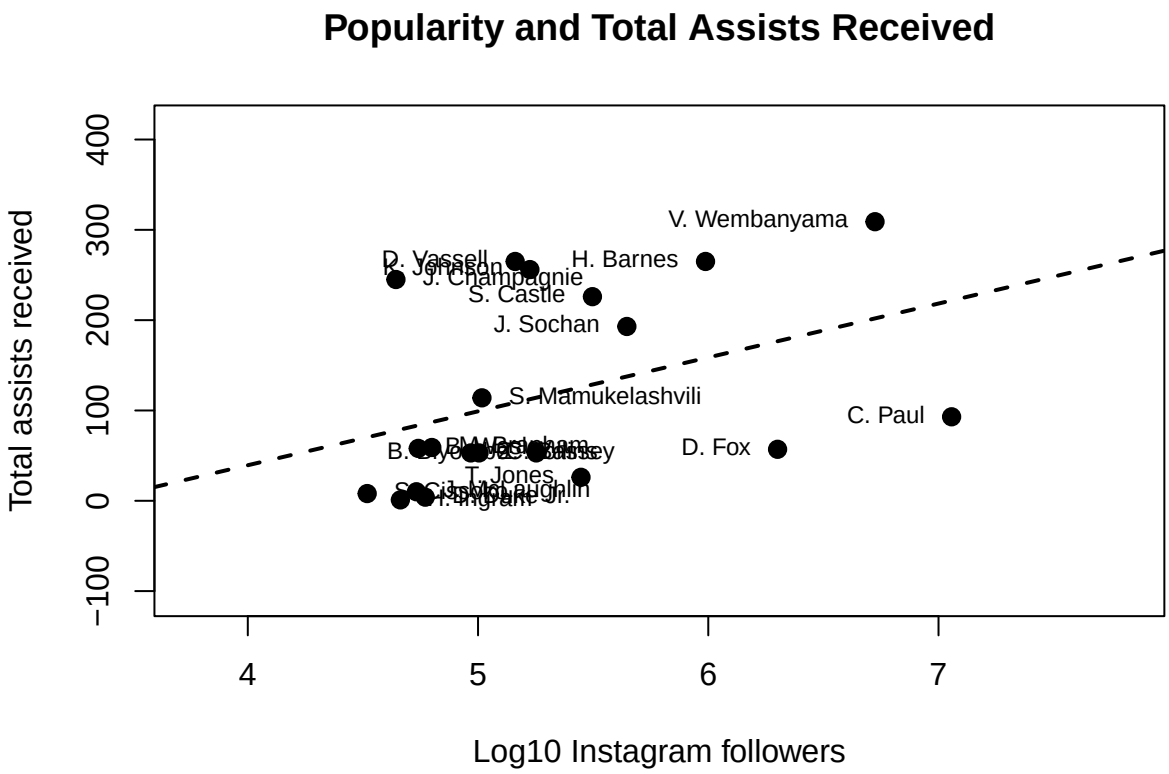
```
## Empirical one-sided p-value: 0.01
```

```r
cat("Empirical two-sided p-value:", round(p_two_sided, 3), "\n")
```

```
## Empirical two-sided p-value: 0.016
```

```r
# Less zoomed scatter plot
short_labels <- sapply(strsplit(as.character(additional_df$player), " "), function(x) {
  if (length(x) >= 2) {
    paste0(substr(x[1], 1, 1), ". ", paste(x[-1], collapse = " "))
  } else {
    x[1]
  }
})

x <- log10(additional_df$followers)
```

```r
y <- additional_df$assists_received

# Bigger margins around the cloud of points
x_pad <- diff(range(x)) * 0.30
y_pad <- diff(range(y)) * 0.35

label_pos <- ifelse(x < median(x), 4, 2)

plot(
  x, y,
  pch = 19,
  cex = 1.2,
  xlim = c(min(x) - x_pad, max(x) + x_pad),
  ylim = c(min(y) - y_pad, max(y) + y_pad),
  xlab = "Log10 Instagram followers",
  ylab = "Total assists received",
  main = "Popularity and Total Assists Received"
)

text(
  x, y,
  labels = short_labels,
  pos = label_pos,
  offset = 0.7,
  cex = 0.75
)

abline(lm(y ~ x), lty = 2, lwd = 2)
```


Popularity and Total Assists Received
Total assists received
Log10 Instagram followers
400
300
200
100
0
−100
4
5
6
7
V. Wembanyama
D. Vassell
H. Barnes
J. Champagnie
S. Castle
J. Sochan
S. Mamukelashvili
C. Paul
D. Fox
T. Jones